

Zeitenwende

Showing in Numbers How the 2022 Russian
Invasion of Ukraine Began an Arms
Reshuffling Across Europe

Clayton Greenwell



Explanation of Project



4 Questions

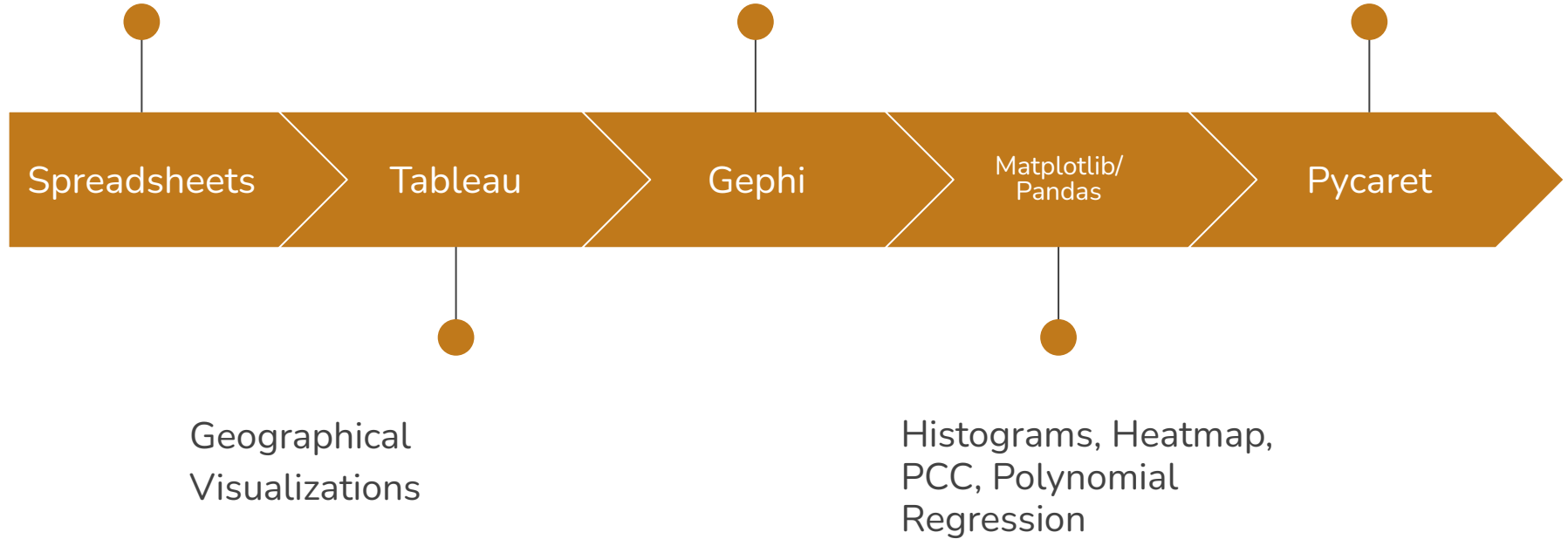
1. Where have European arms exports and imports been most concentrated during the post-invasion period?
2. How did the top-supplier network change between pre-invasion and post-invasion?
3. Which variables were most predictive of military expenditure as a share of GDP, United States top-supplier status, and NATO membership?
4. What European country groupings emerged from clustering with dimensionality reduction?

Data Science Tools Used

Data cleaning (linear interpolation)

Top Supplier Network Analysis

Machine Learning:
Regression, Classification,
Clustering with PCA



Literature Review



- 2014 annexation of Crimea
- Europe has issues with grandiose commitments not leading up to reality
- Only 18% of EU defense investment is done collaboratively

- 1990s & 2000s NATO expansion
- Free riding
- Finland & Sweden

- For every 100 kilometers that a NATO member's capital is located farther from Moscow, their average yearly military expenditure as a percentage of GDP has an associated decrease of 0.02 percentage points. (Knezović & Tkalec, 2025)



- GPR Index

- Keynesian vs.
Neoliberal
Economic Theory

Data Used



Table 1. Data sources

Source	Contribution to the dataset
SIPRI Arms Transfers Database	Arms import TIV units, arms export TIV units, top supplier country.
SIPRI Arms Industry Database	Country representation among the SIPRI Top 100 defense companies.
SIPRI Military Expenditure Database	Military expenditure as a share of GDP, per -capita expenditure, and expenditure as a share of government spending.
Geopolitical Risk Index	Country-level geopolitical-risk indicator from Caldara and Iacoviello (2022).
Google Maps	Aerial distance <u>in</u> kilometers between each national capital and Moscow.

Note. SIPRI = Stockholm International Peace Research Institute; TIV = Trend Indicator Value.



More on Data

- TIV - standardizes production costs by evaluating raw capability and resources required to build hardware.
- Definition of Europe
- Linear Interpolation

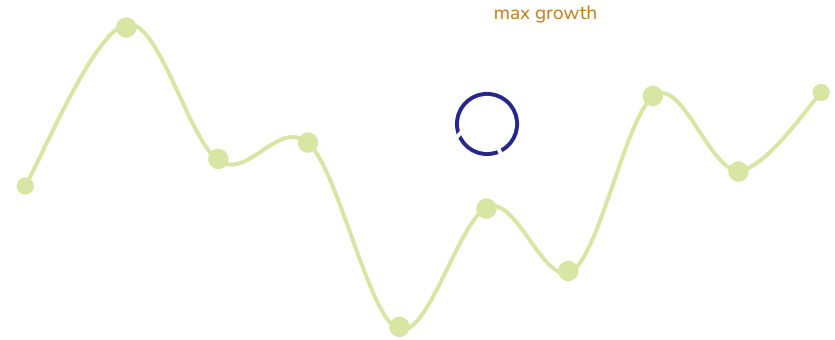
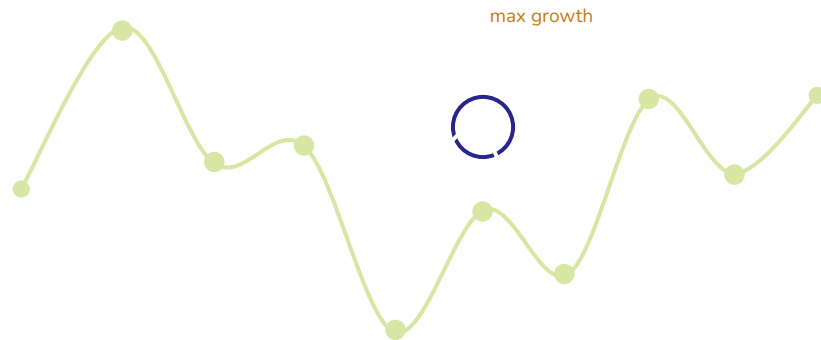


Tableau: Import and Export Maps



Tableau Links

- Total exports of Germany, UK, France, Italy: 7.47 billion TIV units
- <https://public.tableau.com/app/profile/clayton.greenwell/viz/EuropeAverageAnnualArmsExportTIVs20222025/Sheet1>
- <https://public.tableau.com/app/profile/clayton.greenwell/viz/EuropeAverageAnnualArmsImportTIVs2022-2025/Sheet1>

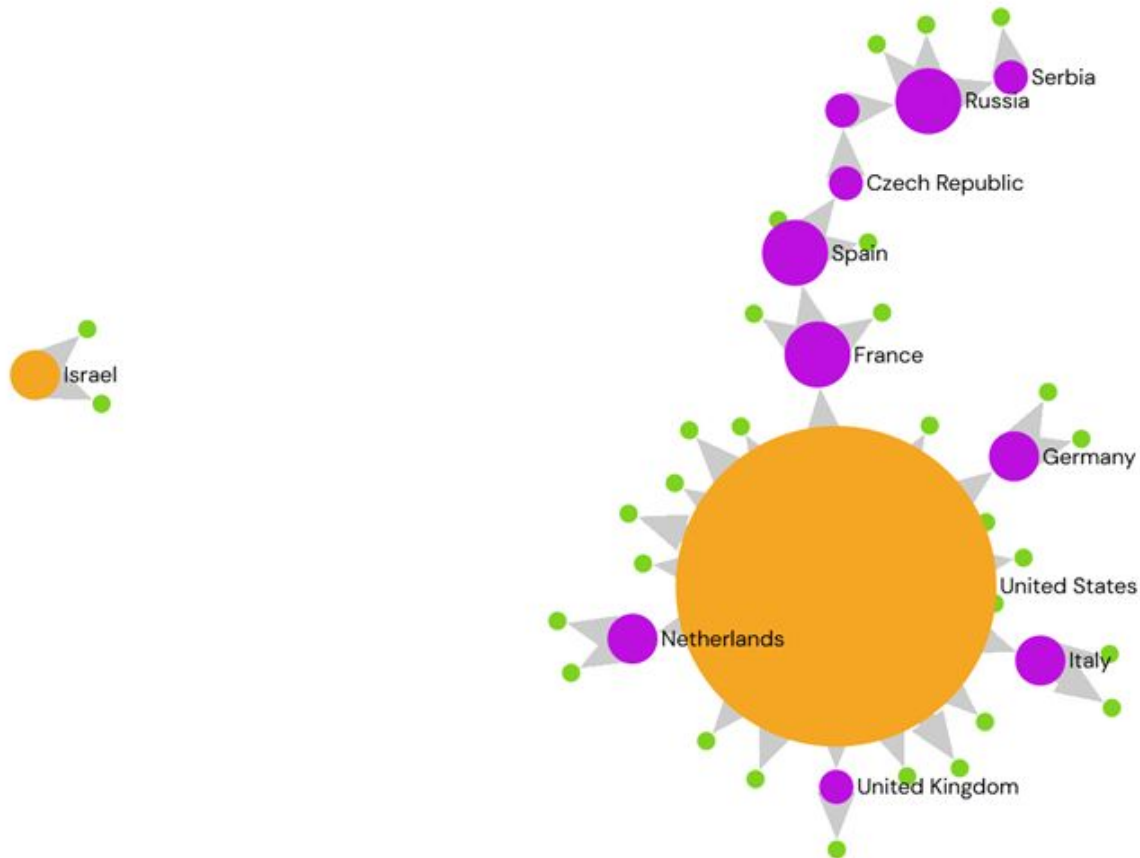




Gephi: Top Supplier Network Analysis

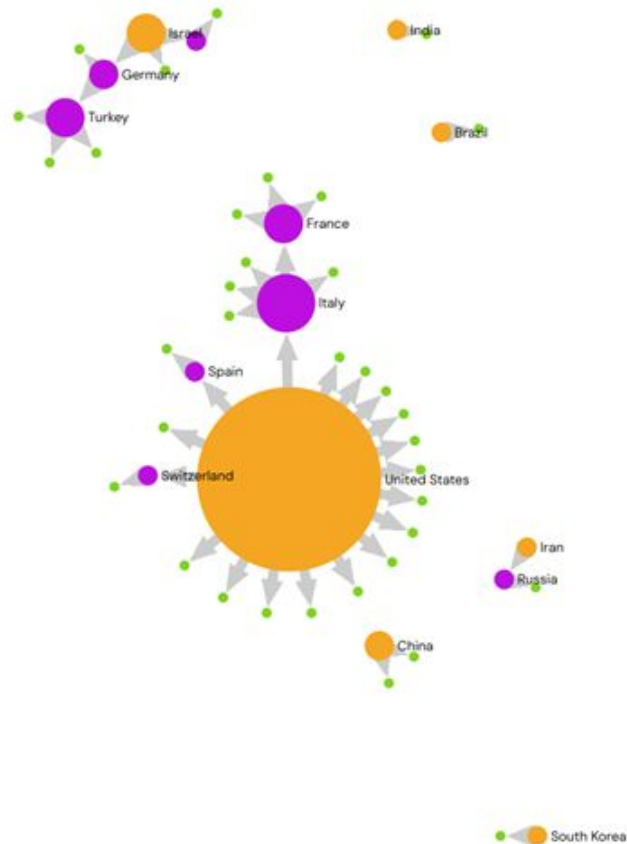
Top Arms Supplier Relationships Before Invasion (2018-2021)

- Recipient
- Supplier and recipient
- Supplier



Top Arms Supplier Relationships After Invasion (2022-2025)

- Recipient
- Supplier and recipient
- Supplier





Regression Model: Military Expenditure as % of GDP

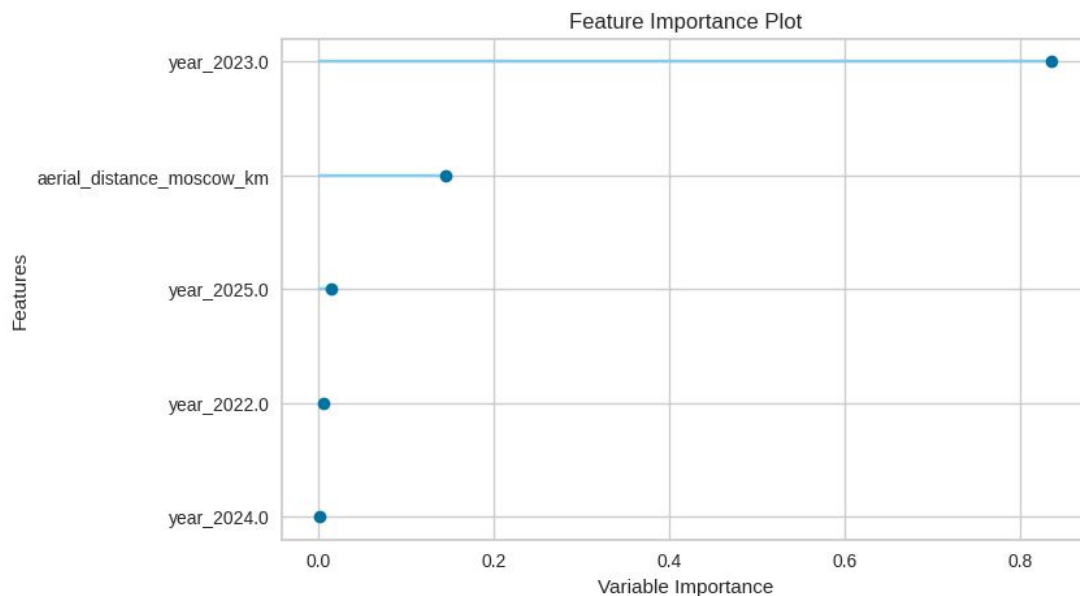


Table 2. Reported performance of the regression model

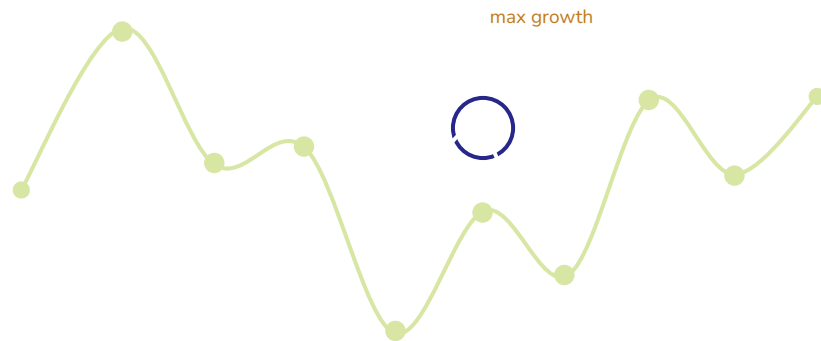
Target	Selected model	Reported metrics	Most important reported features
Military Expenditure as a Percentage of GDP	Extreme Gradient Boosting	MAE = 0.0048; MSE = 0.0001; RMSE = 0.0084; R-squared = 0.82	Calendar year 2023 (>0.80); aerial distance from Moscow (~0.15)

Note. An MAE of 0.0048 corresponds to approximately 0.48 percentage points of GDP.



Why Extreme Gradient Boosting Succeeded

- Boosted tree structured models are better at modeling nonlinear relationships, as well as working with the necessary inclusion of outliers, (Ukraine and Russia) whereas other regression models could easily be pulled astray
- Small dataset





Classification Models: United States Top-Supplier Status & NATO Membership

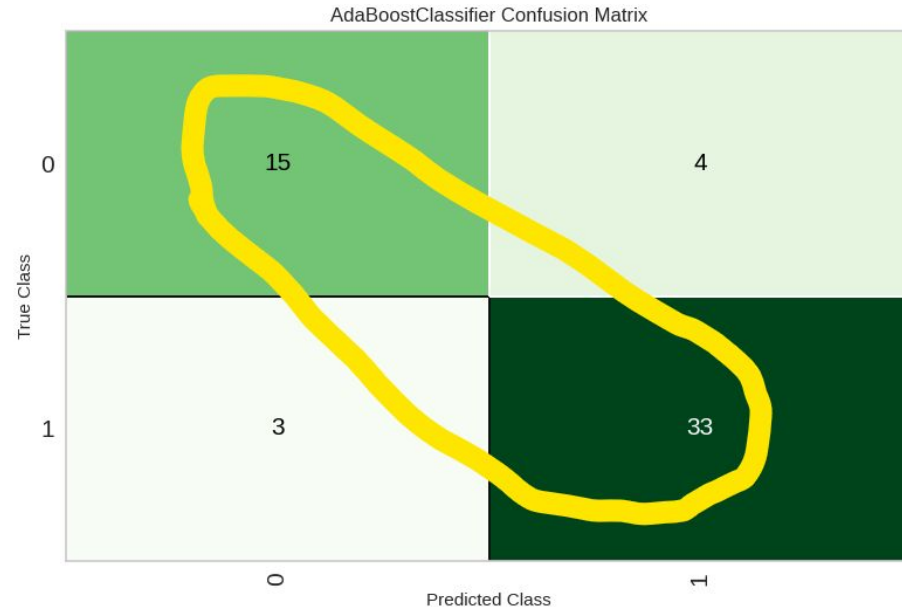
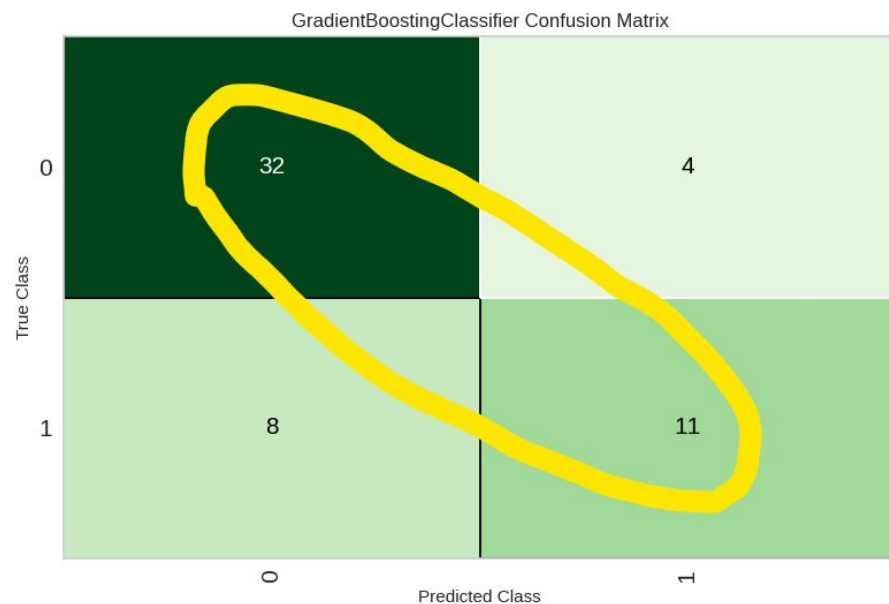


Table 3. Reported performance of the two classification models

Target	Selected model	Reported metrics	Most important reported features
United States as Top Supplier	Gradient Boosting	Accuracy = 0.74; AUC = 0.81; recall = 0.58; precision = 0.61; F1 = 0.58	imports (0.18); military expenditure per capita (~0.14)
NATO Membership	AdaBoost	Accuracy = 0.90; AUC = 0.95; recall = 0.89; precision = 0.96; F1 = 0.92	Aerial distance from Moscow (~0.16); military expenditure per capita (~0.23)

Note. Tuning increased AUC for the United States top-supplier model but reduced accuracy and precision. Tuning did not improve the NATO membership model.

Clustering with PCA



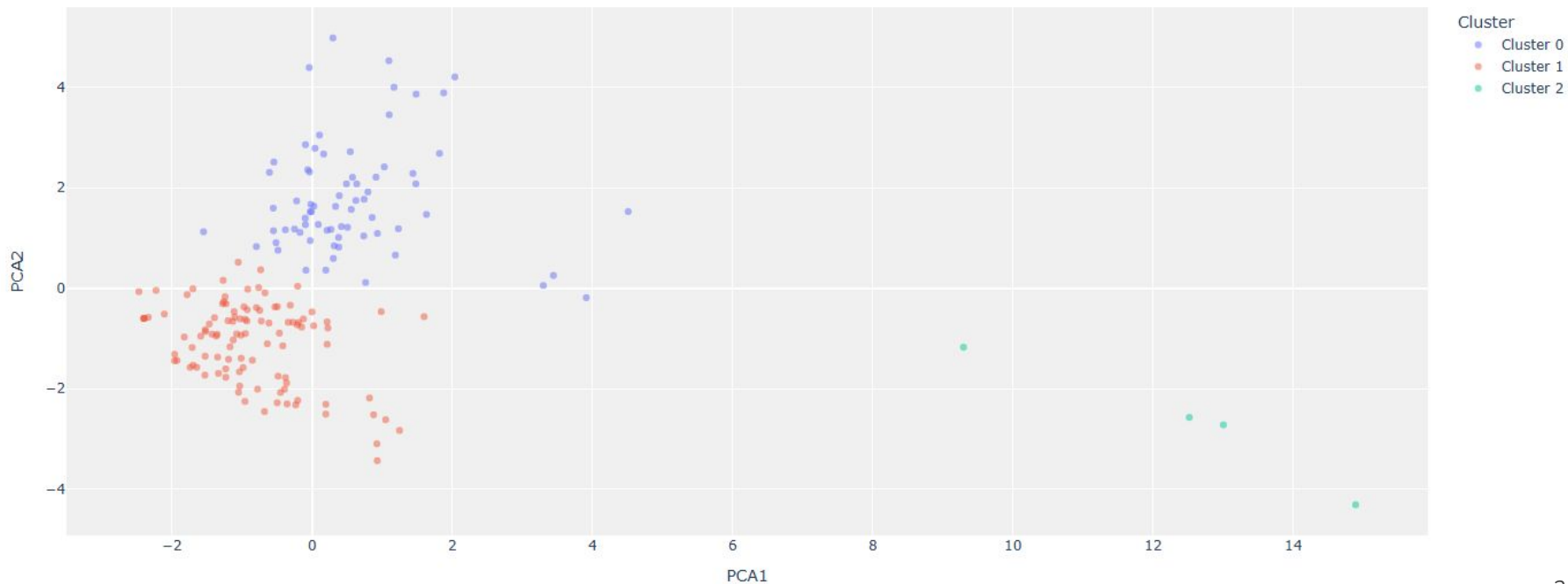
Table 4. Summary of clustering results

Dimensionality Reduction	Algorithms	Clusters and score	Interpretation of clusters
No dimensionality reduction	K-means; hierarchical agglomerative	Best silhouette = 0.22 (K-means)	Weak and difficult-to-interpret structure.
Linear PCA, 2 components	K-means; hierarchical agglomerative	3 clusters; best silhouette = 0.54 (K-means)	Major import/export powers; lower-spending countries; Ukraine as a separate outlier cluster.
Nonlinear kernel PCA, 2 components	K-means; hierarchical agglomerative	3 clusters; best silhouette = 0.65 (tie)	A compact low-import group; a NATO/United States-supplied group; a mixed group containing several major military powers.



K-means Clustering with Linear PCA

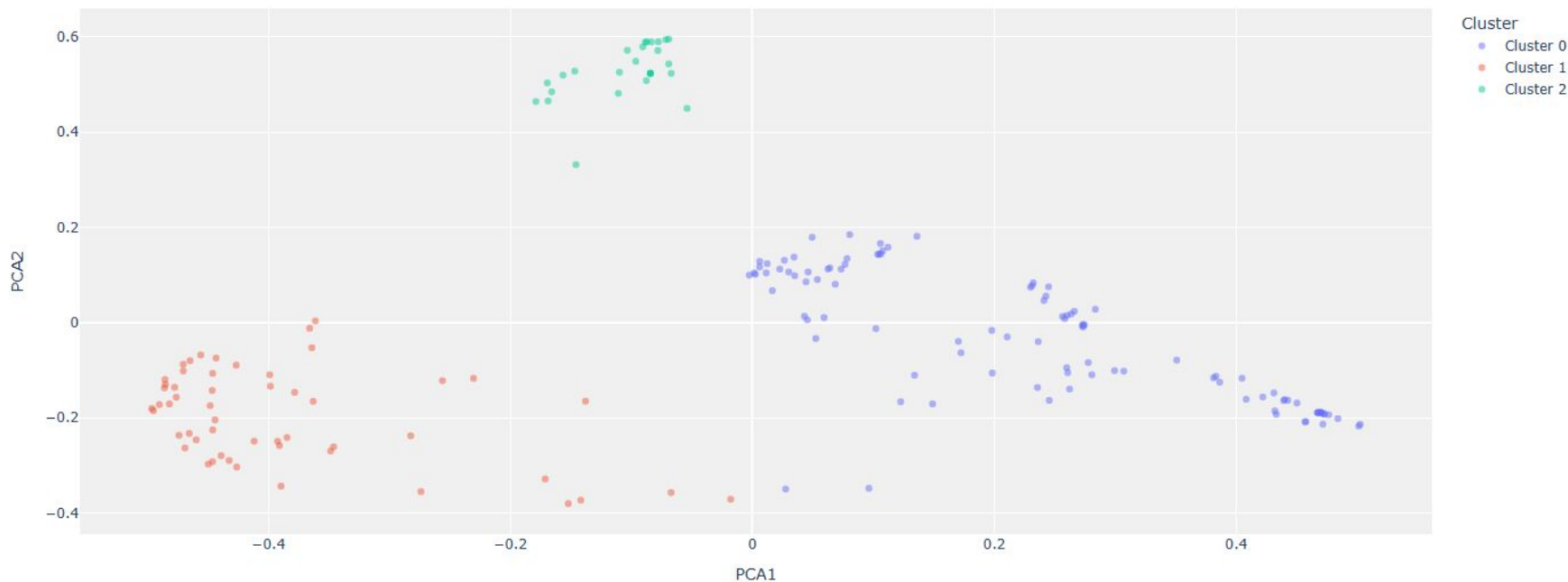
2D Cluster PCA Plot





Agglomerative Clustering with Nonlinear Kernel PCA

2D Cluster PCA Plot





Circling Back to Our 4 Questions



4 Questions

Question 1

Where have European arms exports and imports been most concentrated during the post-invasion period?

Exports: Germany, Italy, UK, France

Imports: Poland, Ukraine

Question 2

How did the top-supplier network change between pre-invasion and post-invasion?

Connected →
Fragmentation



4 Questions

Question 3

Which variables were most predictive of military expenditure as a share of GDP, United States top-supplier status, and NATO membership?

- Calendar Year 2023, distance from Moscow
- ME per capita, arms imports
- ME per capita, distance from Moscow

Question 4

What European country groupings emerged from clustering with dimensionality reduction?

- Low-import group
- Major import/export power group
- Ukraine outlier



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